

Организация сети до удаленных филиалов компании

Network Engineer.
Professional

Меня хорошо видно
& слышно?



Защита проекта

Тема: Организация сети до удаленных филиалов компании

Антониади Федор

ПАО МТС, Отдел контроля сервисов
транспортной сети, ведущий инженер



План защиты

Цель и задачи проекта

Какие технологии
использовались

Что получилось

Выводы

Вопросы и рекомендации

Цель и задачи проекта

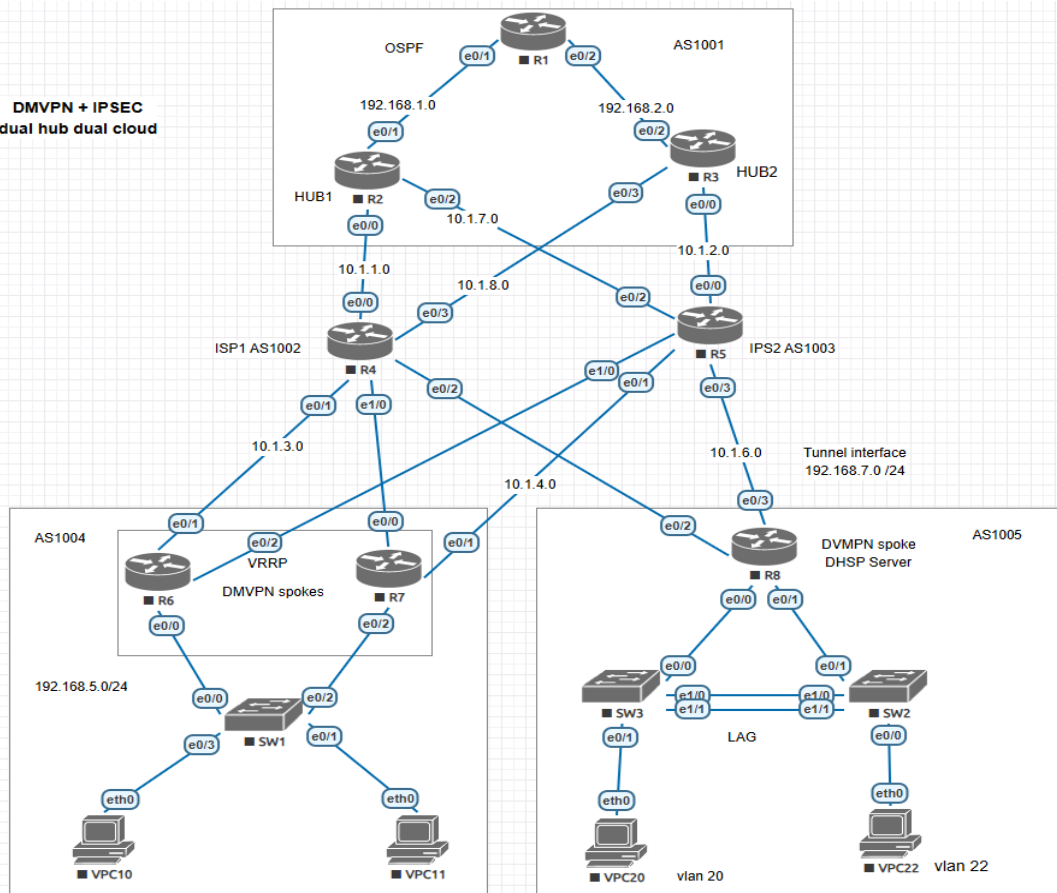
Цель проекта: Создание отказоустойчивой DMVPN (Dynamic Multipoint Virtual Private Network) с архитектурой dual hub dual cloud

Основные задачи, которые необходимо было выполнить для реализации проекта и примененные технологии для их осуществления :

- Создание ip связности между маршрутизаторами с помощью протокола eBGP
- Создание ip связности в офисных сетях между VPC и роутерами
- Отказоустойчивость спокв в одном из филиалов обеспечивается протоколом VRRP (Virtual Router Redundancy Protocol)
- Построение DMVPN между споквами с двумя хабами и двумя сервис-провайдерами
- Шифрование трафика VPN туннелей
- Маршрутизация между офисными сетями филиалов построена на протоколе OSPF

Схема организации связи

DMVPN + IPSEC
dual hub dual cloud



ip план

Equip	Port	Address	Mask	Description	Loopback	Vlan	Vlanif			
			All /24							
R1	E0/1	192.168.1.1		to R2	10.0.0.1/32					
R1	E0/2	192.168.2.1		to R3						
R2	E0/0	10.1.1.2		to R4	10.0.0.2/32					
R2	E0/1	192.168.1.2		to R1						
R2	E0/2	10.1.7.2		to R5						
R3	E0/0	10.1.2.3		to R4	10.0.0.3/32					
R3	E0/2	192.168.2.3		to R1						
R3	E0/3	10.1.8.3		to R4						
R4	E0/0	10.1.1.4		to R2	10.0.0.4/32					
R4	E0/1	10.1.3.4		to R6						
R4	E0/2	10.1.5.4		to R8						
R4	E0/3	10.1.8.3		to R3						
R4	E1/0	10.1.10.4		to R7						
R5	E0/0	10.1.2.5		to R3	10.0.0.5/32					
R5	E0/1	10.1.4.5		to R7						
R5	E0/3	10.1.6.5		to R8						
R5	E0/2	10.1.7.5		to R2						
R5	E1/0	10.1.9.5		to R6						
R6	E0/0	192.168.5.6		to S1	10.0.0.6/32			VRRP 1 (priority 200) 192.168.5.2		
R6	E0/1	10.1.3.6		to R4				VRRP 2 (priority 100) 192.168.5.3		
R6	E0/2	10.1.9.6		to R5						
R7	E0/1	10.1.4.7		to R5	10.0.0.7/32			VRRP 1 (priority 100) 192.168.5.2		
R7	E0/2	192.168.5.7		to S1				VRRP 2 (priority 200) 192.168.5.3		
R7	E0/0	10.1.10.7		to R4						
R8	E0/2	10.1.5.8		to R4						
R8	E0/3	10.1.6.8		to R5						
R8	E0/0.20	192.168.3.1		to S2	10.0.0.8/32			dhcp pool VPC10 Network 192.168.3.0 /24		
R8	E0/1.22	192.168.4.1						dhcp pool VPC22 Network 192.168.4.0 /24		
SW1	E0/0			to R6	10.0.0.10/32					
SW1	E0/2			to R7						
SW1	E0/3			to VPC10						
SW1	E0/1			to VPC11						
SW2	E0/0			to VPC22	10.0.0.20			interface Vlan22 192.168.4.22/24		
SW2	E0/1			to R8						
SW2	E1/0			Port-channel1						
SW2	E1/1			Port-channel1						
SW3	E0/0			to R8	10.0.0.30					
SW3	E0/1			to VPC20				interface Vlan20 192.168.3.30/24		
SW3	E1/0			Port-channel1						
SW3	E1/1			Port-channel1						
VPC10	E0	192.168.5.10		to S1						
VPC11	E0	192.168.5.11		to S1						
VPC20	E0	192.168.3.2		to S2		20				
VPC22	E0	192.168.4.2		to S2		22				

Настройка eBGP на примере пары R2-R4

```
R4#sh run | sec bgp
router bgp 1002
  bgp log-neighbor-changes
  network 10.0.0.4 mask 255.255.255.255
  network 10.1.1.0 mask 255.255.255.0
  network 10.1.3.0 mask 255.255.255.0
  network 10.1.5.0 mask 255.255.255.0
  network 10.1.8.0 mask 255.255.255.0
  network 10.1.10.0 mask 255.255.255.0
  neighbor 10.1.1.2 remote-as 1001
  neighbor 10.1.3.6 remote-as 1004
  neighbor 10.1.5.8 remote-as 1005
  neighbor 10.1.8.3 remote-as 1001
  neighbor 10.1.10.7 remote-as 1004
```

```
R2#sh run | sec bgp
router bgp 1001
  bgp log-neighbor-changes
  network 10.1.1.2 mask 255.255.255.255
  network 10.1.7.0 mask 255.255.255.0
  neighbor 10.1.1.4 remote-as 1002
  neighbor 10.1.7.5 remote-as 1003
```

Проверка bgp соседства

```
R4#sh bgp ipv4 uni summary
BGP router identifier 10.0.0.4, local AS number 1002
BGP table version is 15, main routing table version 15
12 network entries using 1680 bytes of memory
40 path entries using 3200 bytes of memory
7/5 BGP path/bestpath attribute entries using 1008 bytes of memory
6 BGP AS-PATH entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6032 total bytes of memory
BGP activity 12/0 prefixes, 40/0 paths, scan interval 60 secs

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
10.1.1.2      4        1001      71      73      15    0    0 01:00:19      6
10.1.3.6      4        1004      71      74      15    0    0 01:00:10      7
10.1.5.8      4        1005      72      73      15    0    0 00:59:59      7
10.1.8.3      4        1001      71      73      15    0    0 00:59:38      7
10.1.10.7     4        1004      72      73      15    0    0 01:00:06      7
```


Настройка сети в офисе правый

На R8 создаем сабинтерфейсы для vlan 20 и 22

На R8 конфигурируем DHCP сервер для VPC

На свичах пробрасываем вланы для PC и создаем агрегированный линк

Проверяем получение ip адреса на VPC20

```
interface Ethernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.3.1 255.255.255.0
!
interface Ethernet0/1
 no ip address
!
interface Ethernet0/1.22
 encapsulation dot1Q 22
 ip address 192.168.4.1 255.255.255.0
!
interface Ethernet0/2
 description to R4
 ip address 10.1.5.8 255.255.255.0
!
interface Ethernet0/3
 description to R5
 ip address 10.1.6.8 255.255.255.0
```

```
ip dhcp excluded-address 192.168.3.1
ip dhcp excluded-address 192.168.4.1
!
ip dhcp pool VPC22
 network 192.168.4.0 255.255.255.0
 default-router 192.168.4.1
!
ip dhcp pool VPC20
 network 192.168.3.0 255.255.255.0
 default-router 192.168.3.1
```

SW3

```
!
interface Port-channel1
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/0
 switchport trunk allowed vlan 20
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/1
 switchport access vlan 20
 switchport mode access
!
interface Ethernet0/2
!
interface Ethernet0/3
!
interface Ethernet1/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-group 1 mode active
!
interface Ethernet1/1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-group 1 mode active
```

```
VPCS> ip dhcp -r
DDORA IP 192.168.3.2/24 GW 192.168.3.1
VPCS> █
```

Настройка VRRP

 R6

```
interface Ethernet0/0
description to S1
ip address 192.168.5.6 255.255.255.0
vrrp 1 ip 192.168.5.2
vrrp 1 priority 200
vrrp 2 ip 192.168.5.3
```

 R7

```
interface Ethernet0/2
description to S1
ip address 192.168.5.7 255.255.255.0
vrrp 1 ip 192.168.5.2
vrrp 2 ip 192.168.5.3
vrrp 2 priority 200
```

```
R7# show vrrp
Ethernet0/2 - Group 1
  State is Backup
  Virtual IP address is 192.168.5.2
  Virtual MAC address is 0000.5e00.0101
  Advertisement interval is 1.000 sec
  Preemption enabled
  Priority is 100
  Master Router is 192.168.5.6, priority is 200
  Master Advertisement interval is 1.000 sec
  Master Down interval is 3.609 sec (expires in 3.347 sec)

Ethernet0/2 - Group 2
  State is Master
  Virtual IP address is 192.168.5.3
  Virtual MAC address is 0000.5e00.0102
  Advertisement interval is 1.000 sec
  Preemption enabled
  Priority is 200
  Master Router is 192.168.5.7 (local), priority is 200
  Master Advertisement interval is 1.000 sec
  Master Down interval is 3.218 sec
```

 VPC10

```
VPCS> ping 192.168.5.6
```

```
host (192.168.5.6) not reachable
```

```
VPCS> ping 192.168.5.2
```

```
84 bytes from 192.168.5.2 icmp_seq=1 ttl=255 time=1.220 ms
84 bytes from 192.168.5.2 icmp_seq=2 ttl=255 time=1.961 ms
84 bytes from 192.168.5.2 icmp_seq=3 ttl=255 time=2.200 ms
84 bytes from 192.168.5.2 icmp_seq=4 ttl=255 time=1.775 ms
84 bytes from 192.168.5.2 icmp_seq=5 ttl=255 time=1.558 ms
```

```
VPCS> ping 192.168.5.3
```

```
84 bytes from 192.168.5.3 icmp_seq=1 ttl=255 time=2.237 ms
84 bytes from 192.168.5.3 icmp_seq=2 ttl=255 time=1.776 ms
84 bytes from 192.168.5.3 icmp_seq=3 ttl=255 time=1.926 ms
84 bytes from 192.168.5.3 icmp_seq=4 ttl=255 time=1.476 ms
84 bytes from 192.168.5.3 icmp_seq=5 ttl=255 time=1.786 ms
```

Конфигурация DMVPN

```
R2#sh run | sec Tunnel1
interface Tunnel1
 ip address 192.168.7.2 255.255.255.0
 no ip redirects
 ip nhrp map multicast dynamic
 ip nhrp network-id 1
 ip nhrp redirect
 ip ospf network broadcast
 ip ospf priority 100
 tunnel source Ethernet0/0
 tunnel mode gre multipoint
 tunnel protection ipsec profile VTI
```

```
R3# sh run | sec Tunnel1
interface Tunnel1
 ip address 192.168.7.3 255.255.255.0
 no ip redirects
 ip nhrp map multicast dynamic
 ip nhrp network-id 1
 ip nhrp redirect
 ip ospf network broadcast
 ip ospf priority 100
 tunnel source Ethernet0/0
 tunnel mode gre multipoint
 tunnel protection ipsec profile VTI
```

```
R6#sh run | sec Tunnel1
interface Tunnel1
 ip address 192.168.7.6 255.255.255.0
 no ip redirects
 ip nhrp map multicast 10.1.1.2
 ip nhrp map 192.168.7.2 10.1.1.2
 ip nhrp map 192.168.7.3 10.1.2.3
 ip nhrp map multicast 10.1.2.3
 ip nhrp network-id 1
 ip nhrp nhs 192.168.7.2
 ip nhrp nhs 192.168.7.3
 ip nhrp registration no-unique
 ip nhrp shortcut
 ip ospf network broadcast
 ip ospf priority 0
 tunnel source Ethernet0/1
 tunnel mode gre multipoint
 tunnel protection ipsec profile VTI
```

```
R7#sh run | sec Tunnel1
interface Tunnel1
 ip address 192.168.7.7 255.255.255.0
 no ip redirects
 ip nhrp map multicast 10.1.2.3
 ip nhrp map 192.168.7.3 10.1.2.3
 ip nhrp map multicast 10.1.1.2
 ip nhrp map 192.168.7.2 10.1.1.2
 ip nhrp network-id 1
 ip nhrp nhs 192.168.7.3
 ip nhrp nhs 192.168.7.2
 ip nhrp registration no-unique
 ip nhrp shortcut
 ip ospf network broadcast
 ip ospf priority 0
 tunnel source Ethernet0/1
 tunnel mode gre multipoint
 tunnel protection ipsec profile VTI
```

```
R8#sh run | sec Tunnel1
interface Tunnel1
 ip address 192.168.7.8 255.255.255.0
 no ip redirects
 ip nhrp map multicast 10.1.1.2
 ip nhrp map 192.168.7.2 10.1.1.2
 ip nhrp map 192.168.7.3 10.1.2.3
 ip nhrp map multicast 10.1.2.3
 ip nhrp network-id 1
 ip nhrp nhs 192.168.7.2
 ip nhrp nhs 192.168.7.3
 ip nhrp registration no-unique
 ip nhrp shortcut
 ip ospf network broadcast
 ip ospf priority 0
 tunnel source Ethernet0/3
 tunnel mode gre multipoint
 tunnel protection ipsec profile VTI
```

Проверка DMVPN на хабе и на споке

```
R6#sh ip nhrp br
  Target          Via          NBMA          Mode    Intfc    Claimed
  192.168.7.2/32  192.168.7.2  10.1.1.2      static  Tul      <    >
  192.168.7.3/32  192.168.7.3  10.1.2.3      static  Tul      <    >
  192.168.7.6/32  192.168.7.6  10.1.3.6      dynamic Tul      <    >
  192.168.7.7/32  192.168.7.7  10.1.4.7      dynamic Tul      <    >
  192.168.7.8/32  192.168.7.8  10.1.6.8      dynamic Tul      <    >
```

```
R2#sh dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel

=====
```

```
Interface: Tunnell, IPv4 NHRP Details
Type:Hub, NHRP Peers:3,
```

# Ent	Peer NBMA Addr	Peer Tunnel Add	State	UpDn Tm	Attrb
1	10.1.3.6	192.168.7.6	UP	00:50:35	D
1	10.1.4.7	192.168.7.7	UP	02:54:16	D
1	10.1.6.8	192.168.7.8	UP	02:54:17	D

```
R7>ping 192.168.7.6
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.7.6, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 6/7/8 ms
R7>ping 192.168.7.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.7.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 5/12/18 ms
R7>ping 192.168.7.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.7.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 7/9/13 ms
R7>ping 192.168.7.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.7.3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 7/7/8 ms
```

Настройка ipsec

```
R8#sh run | sec crypto
crypto isakmp policy 10
  encr aes
  hash sha256
  authentication pre-share
  group 5
crypto isakmp key DMVPN_OTUS address 0.0.0.0
crypto ipsec transform-set VTI esp-aes esp-sha-hmac
  mode transport
crypto ipsec profile VTI
  set transform-set VTI
```

```
R8#ping 192.168.7.6
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.7.6, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 6/7/8 ms
R8#show crypto ipsec sa peer 10.1.3.6

interface: Tunnell
  Crypto map tag: Tunnell-head-0, local addr 10.1.6.8

protected vrf: (none)
local ident (addr/mask/prot/port): (10.1.6.8/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (10.1.3.6/255.255.255.255/47/0)
current_peer 10.1.3.6 port 500
  PERMIT, flags={origin is acl,}
  #pkts encaps: 17, #pkts encrypt: 17, #pkts digest: 17
  #pkts decaps: 16, #pkts decrypt: 16, #pkts verify: 16
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
```

```
R8#show crypto session
Crypto session current status
```

```
Interface: Tunnell
Session status: UP-ACTIVE
Peer: 10.1.4.7 port 500
  Session ID: 0
  IKEv1 SA: local 10.1.6.8/500 remote 10.1.4.7/500 Active
  Session ID: 0
  IKEv1 SA: local 10.1.6.8/500 remote 10.1.4.7/500 Active
  IPSEC FLOW: permit 47 host 10.1.6.8 host 10.1.4.7
    Active SAs: 2, origin: crypto map
```

```
Interface: Tunnell
Session status: UP-ACTIVE
Peer: 10.1.3.6 port 500
  Session ID: 0
  IKEv1 SA: local 10.1.6.8/500 remote 10.1.3.6/500 Active
  Session ID: 0
  IKEv1 SA: local 10.1.6.8/500 remote 10.1.3.6/500 Active
  IPSEC FLOW: permit 47 host 10.1.6.8 host 10.1.3.6
    Active SAs: 4, origin: crypto map
```

```
Interface: Tunnell
Session status: UP-ACTIVE
Peer: 10.1.2.3 port 500
  Session ID: 0
  IKEv1 SA: local 10.1.6.8/500 remote 10.1.2.3/500 Active
  IPSEC FLOW: permit 47 host 10.1.6.8 host 10.1.2.3
    Active SAs: 2, origin: crypto map
```

```
Interface: Tunnell
Session status: UP-ACTIVE
Peer: 10.1.1.2 port 500
  Session ID: 0
  IKEv1 SA: local 10.1.6.8/500 remote 10.1.1.2/500 Active
  IPSEC FLOW: permit 47 host 10.1.6.8 host 10.1.1.2
    Active SAs: 2, origin: crypto map
```

Настройка OSPF

```
R2#sh run | sec ospf
ip ospf network broadcast
ip ospf priority 100
router ospf 1
area 10 stub
network 10.0.0.0 0.0.0.255 area 10
network 192.168.1.0 0.0.0.255 area 10
network 192.168.7.0 0.0.0.255 area 0
```

```
R6#sh run | sec ospf
ip ospf network broadcast
ip ospf priority 0
router ospf 1
area 20 stub
network 10.0.0.0 0.0.0.255 area 20
network 192.168.5.0 0.0.0.255 area 20
network 192.168.7.0 0.0.0.255 area 0
```

```
R7>sh ip ospf nei
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.0.0.2	100	2WAY/DROTHER	00:00:32	192.168.7.2	Tunnell
10.0.0.3	100	FULL/DR	00:00:33	192.168.7.3	Tunnell
10.0.0.6	1	FULL/BDR	00:00:31	192.168.5.6	Ethernet0/2

```
R3#sh ip ospf nei
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.0.0.6	0	FULL/DROTHER	00:00:34	192.168.7.6	Tunnell
10.0.0.7	0	FULL/DROTHER	00:00:33	192.168.7.7	Tunnell
10.0.0.8	0	FULL/DROTHER	00:00:33	192.168.7.8	Tunnell
10.0.0.1	1	FULL/BDR	00:00:38	192.168.2.1	Ethernet0/2

```

R2#sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 10.1.1.4 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.1.1.4
      10.0.0.0/8 is variably subnetted, 21 subnets, 2 masks
C      10.0.0.0/24 is directly connected, Loopback0
O      10.0.0.1/32 [110/11] via 192.168.1.1, 00:59:26, Ethernet0/1
L      10.0.0.2/32 is directly connected, Loopback0
O      10.0.0.3/32 [110/21] via 192.168.1.1, 00:59:26, Ethernet0/1
B      10.0.0.4/32 [20/0] via 10.1.1.4, 00:59:12
B      10.0.0.5/32 [20/0] via 10.1.7.5, 00:59:13
O IA   10.0.0.6/32 [110/1001] via 192.168.7.6, 00:16:05, Tunnell
O IA   10.0.0.7/32 [110/1001] via 192.168.7.7, 00:16:05, Tunnell
O IA   10.0.0.8/32 [110/1001] via 192.168.7.8, 00:16:05, Tunnell
C      10.1.1.0/24 is directly connected, Ethernet0/0
L      10.1.1.2/32 is directly connected, Ethernet0/0
B      10.1.2.0/24 [20/0] via 10.1.7.5, 00:59:13
B      10.1.3.0/24 [20/0] via 10.1.1.4, 00:59:12
B      10.1.4.0/24 [20/0] via 10.1.7.5, 00:59:13
B      10.1.5.0/24 [20/0] via 10.1.1.4, 00:59:12
B      10.1.6.0/24 [20/0] via 10.1.7.5, 00:59:13
C      10.1.7.0/24 is directly connected, Ethernet0/2
L      10.1.7.2/32 is directly connected, Ethernet0/2
B      10.1.8.0/24 [20/0] via 10.1.1.4, 00:59:12
B      10.1.9.0/24 [20/0] via 10.1.7.5, 00:59:13
B      10.1.10.0/24 [20/0] via 10.1.1.4, 00:59:12
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/24 is directly connected, Ethernet0/1
L      192.168.1.2/32 is directly connected, Ethernet0/1
O      192.168.2.0/24 [110/20] via 192.168.1.1, 00:59:26, Ethernet0/1
O IA   192.168.3.0/24 [110/1010] via 192.168.7.8, 00:16:05, Tunnell
O IA   192.168.4.0/24 [110/1010] via 192.168.7.8, 00:16:05, Tunnell
O IA   192.168.5.0/24 [110/1010] via 192.168.7.7, 00:16:05, Tunnell
      [110/1010] via 192.168.7.6, 00:16:05, Tunnell
      192.168.7.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.7.0/24 is directly connected, Tunnell
L      192.168.7.2/32 is directly connected, Tunnell

```

```

R8#sh ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

```

Gateway of last resort is 10.1.6.5 to network 0.0.0.0

```

      10.0.0.0/8 is variably subnetted, 20 subnets, 2 masks
O IA   10.0.0.1/32 [110/1011] via 192.168.7.3, 00:23:25, Tunnell
      [110/1011] via 192.168.7.2, 00:23:54, Tunnell
O IA   10.0.0.2/32 [110/1001] via 192.168.7.2, 00:23:54, Tunnell
O IA   10.0.0.3/32 [110/1001] via 192.168.7.3, 00:23:25, Tunnell
O IA   10.0.0.6/32 [110/1001] via 192.168.7.6, 01:06:54, Tunnell
O IA   10.0.0.7/32 [110/1001] via 192.168.7.7, 01:06:44, Tunnell
O IA   192.168.1.0/24 [110/1010] via 192.168.7.2, 00:23:54, Tunnell
O IA   192.168.2.0/24 [110/1010] via 192.168.7.3, 00:23:25, Tunnell
O IA   192.168.5.0/24 [110/1010] via 192.168.7.7, 01:06:44, Tunnell
      [110/1010] via 192.168.7.6, 01:06:54, Tunnell

```

```

O*IA   0.0.0.0/0 [110/11] via 192.168.2.3, 01:13:05, Ethernet0/2
      [110/11] via 192.168.1.2, 01:12:55, Ethernet0/1
      10.0.0.0/32 is subnetted, 6 subnets
C      10.0.0.1 is directly connected, Loopback0
O      10.0.0.2 [110/11] via 192.168.1.2, 01:12:55, Ethernet0/1
O      10.0.0.3 [110/11] via 192.168.2.3, 01:13:05, Ethernet0/2
O IA   10.0.0.6 [110/1011] via 192.168.2.3, 00:29:12, Ethernet0/2
O IA   10.0.0.7 [110/1011] via 192.168.2.3, 00:29:12, Ethernet0/2
O IA   10.0.0.8 [110/1011] via 192.168.2.3, 00:29:12, Ethernet0/2
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/24 is directly connected, Ethernet0/1
L      192.168.1.1/32 is directly connected, Ethernet0/1
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.2.0/24 is directly connected, Ethernet0/2
L      192.168.2.1/32 is directly connected, Ethernet0/2
O IA   192.168.3.0/24 [110/1020] via 192.168.2.3, 00:29:12, Ethernet0/2
O IA   192.168.4.0/24 [110/1020] via 192.168.2.3, 00:29:12, Ethernet0/2
O IA   192.168.5.0/24 [110/1020] via 192.168.2.3, 00:29:12, Ethernet0/2
O IA   192.168.7.0/24 [110/1010] via 192.168.2.3, 01:13:05, Ethernet0/2
      [110/1010] via 192.168.1.2, 01:12:55, Ethernet0/1

```

R1

Проверка ip связности между VPC

```
VPC10
VPCS>
VPCS>
VPCS>
VPCS> sh ip
NAME      : VPCS[1]
IP/MASK    : 192.168.5.10/24
GATEWAY    : 192.168.5.2
DNS        :
MAC        : 00:50:79:66:68:0e
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.4.2

84 bytes from 192.168.4.2 icmp_seq=1 ttl=62 time=5.512 ms
84 bytes from 192.168.4.2 icmp_seq=2 ttl=62 time=4.518 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=62 time=5.674 ms
84 bytes from 192.168.4.2 icmp_seq=4 ttl=62 time=5.587 ms
84 bytes from 192.168.4.2 icmp_seq=5 ttl=62 time=4.753 ms

VPCS>
```

```
VPC11
VPCS>
VPCS>
VPCS>
VPCS>
VPCS> sh ip
NAME      : VPCS[1]
IP/MASK    : 192.168.5.11/24
GATEWAY    : 192.168.5.3
DNS        :
MAC        : 00:50:79:66:68:0f
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> trace 192.168.3.2
trace to 192.168.3.2, 8 hops max, press Ctrl+C to stop
 1  192.168.5.7   1.253 ms  0.930 ms  1.004 ms
 2  192.168.7.8   2.646 ms  2.548 ms  2.808 ms
 3  *192.168.3.2   3.965 ms  (ICMP type:3, code:3, Destination port unreachable)

VPCS>
```

```
VPC20
VPCS>
VPCS>
VPCS> sh ip
NAME      : VPCS[1]
IP/MASK    : 192.168.3.2/24
GATEWAY    : 192.168.3.1
DNS        :
DHCP SERVER : 192.168.3.1
DHCP LEASE  : 85821, 86400/43200/75600
MAC        : 00:50:79:66:68:10
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> trace 192.168.5.10
trace to 192.168.5.10, 8 hops max, press Ctrl+C to stop
 1  192.168.3.1   1.689 ms  1.811 ms  1.910 ms
 2  192.168.7.7   3.150 ms  2.795 ms  2.691 ms
 3  *192.168.5.10 5.747 ms  (ICMP type:3, code:3, Destination port unreachable)

VPCS>
```

```
VPC22
VPCS>
VPCS>
VPCS> sh ip
NAME      : VPCS[1]
IP/MASK    : 192.168.4.2/24
GATEWAY    : 192.168.4.1
DNS        :
DHCP SERVER : 192.168.4.1
DHCP LEASE  : 85821, 86400/43200/75600
MAC        : 00:50:79:66:68:11
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.5.11

84 bytes from 192.168.5.11 icmp_seq=1 ttl=62 time=9.442 ms
84 bytes from 192.168.5.11 icmp_seq=2 ttl=62 time=6.608 ms
84 bytes from 192.168.5.11 icmp_seq=3 ttl=62 time=4.619 ms
84 bytes from 192.168.5.11 icmp_seq=4 ttl=62 time=4.458 ms
84 bytes from 192.168.5.11 icmp_seq=5 ttl=62 time=4.711 ms

VPCS>
```



Проверка отказоустойчивости

```
VPC10
VPCS> ping 10.0.0.4
10.0.0.4 icmp_seq=1 timeout
10.0.0.4 icmp_seq=2 timeout
10.0.0.4 icmp_seq=3 timeout
10.0.0.4 icmp_seq=4 timeout
10.0.0.4 icmp_seq=5 timeout

VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS> trace 192.168.4.2
trace to 192.168.4.2, 8 hops max, press Ctrl+C to stop
 1 192.168.5.6  2.807 ms  1.634 ms  2.003 ms
 2 192.168.7.8  2.029 ms  1.769 ms  2.254 ms
 3 *192.168.4.2  2.804 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>

VPC20
VPCS> ping 10.0.0.4
10.0.0.4 icmp_seq=1 timeout
10.0.0.4 icmp_seq=2 timeout
10.0.0.4 icmp_seq=3 timeout
10.0.0.4 icmp_seq=4 timeout
10.0.0.4 icmp_seq=5 timeout

VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS> trace 192.168.5.10
trace to 192.168.5.10, 8 hops max, press Ctrl+C to stop
 1 192.168.3.1  1.694 ms  1.823 ms  1.771 ms
 2 192.168.7.7  46.365 ms  3.856 ms  6.894 ms
 3 *192.168.5.10  4.749 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>

VPC11
10.0.0.4 icmp_seq=1 timeout
10.0.0.4 icmp_seq=2 timeout
10.0.0.4 icmp_seq=3 timeout
10.0.0.4 icmp_seq=4 timeout
10.0.0.4 icmp_seq=5 timeout

VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS> trace 192.168.3.2
trace to 192.168.3.2, 8 hops max, press Ctrl+C to stop
 1 192.168.5.7  1.248 ms  1.346 ms  1.457 ms
 2 192.168.7.8  10.231 ms  4.291 ms  3.188 ms
 3 *192.168.3.2  5.763 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>

VPC22
VPCS> ping 10.0.0.4
10.0.0.4 icmp_seq=1 timeout
10.0.0.4 icmp_seq=2 timeout
10.0.0.4 icmp_seq=3 timeout
10.0.0.4 icmp_seq=4 timeout
10.0.0.4 icmp_seq=5 timeout

VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS> trace 192.168.5.11
trace to 192.168.5.11, 8 hops max, press Ctrl+C to stop
 1 192.168.4.1  1.890 ms  1.832 ms  1.468 ms
 2 192.168.7.7  4.079 ms  4.439 ms  4.704 ms
 3 *192.168.5.11  8.779 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>
```



```
VPC10
2 192.168.7.8 2.029 ms 1.769 ms 2.254 ms
3 *192.168.4.2 2.804 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> ping 10.0.0.7

10.0.0.7 icmp_seq=1 timeout
10.0.0.7 icmp_seq=2 timeout
10.0.0.7 icmp_seq=3 timeout
10.0.0.7 icmp_seq=4 timeout
10.0.0.7 icmp_seq=5 timeout

VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS> trace 192.168.4.2
trace to 192.168.4.2, 8 hops max, press Ctrl+C to stop
1 192.168.5.6 1.969 ms 2.294 ms 1.588 ms
2 192.168.7.8 4.945 ms 4.960 ms 2.139 ms
3 *192.168.4.2 3.036 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>

VPC20
)
VPCS> ping 10.0.0.7

10.0.0.7 icmp_seq=1 timeout
10.0.0.7 icmp_seq=2 timeout
10.0.0.7 icmp_seq=3 timeout
10.0.0.7 icmp_seq=4 timeout
10.0.0.7 icmp_seq=5 timeout

VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS> trace 192.168.5.10
trace to 192.168.5.10, 8 hops max, press Ctrl+C to stop
1 192.168.3.1 2.428 ms 1.649 ms 2.090 ms
2 192.168.7.6 4.064 ms 4.104 ms 3.182 ms
3 *192.168.5.10 2.470 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>

VPC11
1 192.168.5.7 1.248 ms 1.346 ms 1.457 ms
2 192.168.7.8 10.231 ms 4.291 ms 3.188 ms
3 *192.168.3.2 5.763 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> ping 10.0.0.7

10.0.0.7 icmp_seq=1 timeout
10.0.0.7 icmp_seq=2 timeout
10.0.0.7 icmp_seq=3 timeout
10.0.0.7 icmp_seq=4 timeout
10.0.0.7 icmp_seq=5 timeout

VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS> trace 192.168.3.2
trace to 192.168.3.2, 8 hops max, press Ctrl+C to stop
1 192.168.5.6 2.225 ms 1.709 ms 1.649 ms
2 192.168.7.8 1.813 ms 3.172 ms 4.385 ms
3 *192.168.3.2 3.883 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>

VPC22
)
VPCS> ping 10.0.0.7

10.0.0.7 icmp_seq=1 timeout
10.0.0.7 icmp_seq=2 timeout
10.0.0.7 icmp_seq=3 timeout
10.0.0.7 icmp_seq=4 timeout
10.0.0.7 icmp_seq=5 timeout

VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS> trace 192.168.5.11
trace to 192.168.5.11, 8 hops max, press Ctrl+C to stop
1 192.168.4.1 1.634 ms 0.742 ms 1.155 ms
2 192.168.7.6 3.369 ms 2.747 ms 3.064 ms
3 *192.168.5.11 3.754 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>
```



```
VPC10
VPCS>
VPCS> ping 10.0.0.3

10.0.0.3 icmp_seq=1 timeout
10.0.0.3 icmp_seq=2 timeout
10.0.0.3 icmp_seq=3 timeout
10.0.0.3 icmp_seq=4 timeout
10.0.0.3 icmp_seq=5 timeout

VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS> ping 192.168.4.2

84 bytes from 192.168.4.2 icmp_seq=1 ttl=62 time=5.214 ms
84 bytes from 192.168.4.2 icmp_seq=2 ttl=62 time=6.062 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=62 time=5.042 ms
84 bytes from 192.168.4.2 icmp_seq=4 ttl=62 time=5.065 ms
84 bytes from 192.168.4.2 icmp_seq=5 ttl=62 time=8.003 ms

VPCS> █

VPC20
VPCS>
VPCS> ping 10.0.0.3

10.0.0.3 icmp_seq=1 timeout
10.0.0.3 icmp_seq=2 timeout
10.0.0.3 icmp_seq=3 timeout
10.0.0.3 icmp_seq=4 timeout
10.0.0.3 icmp_seq=5 timeout

VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS> ping 192.168.5.11

84 bytes from 192.168.5.11 icmp_seq=1 ttl=62 time=5.821 ms
84 bytes from 192.168.5.11 icmp_seq=2 ttl=62 time=4.980 ms
84 bytes from 192.168.5.11 icmp_seq=3 ttl=62 time=5.906 ms
84 bytes from 192.168.5.11 icmp_seq=4 ttl=62 time=5.341 ms
84 bytes from 192.168.5.11 icmp_seq=5 ttl=62 time=5.368 ms

VPCS> █

VPC11
VPCS> ping 10.0.0.3

10.0.0.3 icmp_seq=1 timeout
10.0.0.3 icmp_seq=2 timeout
10.0.0.3 icmp_seq=3 timeout
10.0.0.3 icmp_seq=4 timeout
10.0.0.3 icmp_seq=5 timeout

VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS> ping 192.168.3.2

84 bytes from 192.168.3.2 icmp_seq=1 ttl=62 time=5.736 ms
84 bytes from 192.168.3.2 icmp_seq=2 ttl=62 time=5.978 ms
84 bytes from 192.168.3.2 icmp_seq=3 ttl=62 time=5.074 ms
84 bytes from 192.168.3.2 icmp_seq=4 ttl=62 time=7.344 ms
84 bytes from 192.168.3.2 icmp_seq=5 ttl=62 time=5.952 ms

VPCS> █

VPC22
VPCS> ping 10.0.0.3

10.0.0.3 icmp_seq=1 timeout
10.0.0.3 icmp_seq=2 timeout
10.0.0.3 icmp_seq=3 timeout
10.0.0.3 icmp_seq=4 timeout
10.0.0.3 icmp_seq=5 timeout

VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS>
VPCS> ping 192.168.5.10

84 bytes from 192.168.5.10 icmp_seq=1 ttl=62 time=6.973 ms
84 bytes from 192.168.5.10 icmp_seq=2 ttl=62 time=6.951 ms
84 bytes from 192.168.5.10 icmp_seq=3 ttl=62 time=5.465 ms
84 bytes from 192.168.5.10 icmp_seq=4 ttl=62 time=9.868 ms
84 bytes from 192.168.5.10 icmp_seq=5 ttl=62 time=5.494 ms

VPCS> █
```



Выводы

Оцените работу над проектом и ответьте на вопросы:

У вас получилось достичь цели и выполнить все задачи?

Что далось легко, а с чем возникли трудности?

Сколько времени занял проект?

Насколько полезным оказался для вас проект от 1 до 10?

1 = я не научился ничему новому

10 = очень полезно, я получил новый опыт

Остались ли у вас вопросы по проекту?

Как вы планируете развиваться дальше?

Ответьте на вопросы
одногогруппников и
преподавателей и получите
обратную связь на свою
работу

Вопросы и рекомендации

+ если есть
вопросы

— если вопросов нет



Спасибо за внимание!